



Order, Cost & Stability in Diffusion Sampling

A numerical study of exponential integrators for the probability-flow ODE

CSE-402 · Numerical Analysis, Simulation & Modeling · Course Project Proposal · Section A

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01 · Mathematical Formulation & Model I/O

The forward SDE, the probability-flow ODE it induces, and what enters and leaves the score network.

ADDRESSES FEEDBACK 1

02 · Base Paper & The Numerical Core

DPM-Solver read as an exponential integrator, and the three verification gaps this project targets.

03 · Generalizability of Findings

A three-tier validation suite spanning stiff linear, non-linear and empirical image manifolds.

ADDRESSES FEEDBACK 2

04 · Methodology, Solvers & Metrics

Seven solvers on identical ODEs at an equal budget of network evaluations, judged on order, stability and cost.

One-line thesis: diffusion sampling is an initial value problem, so its solvers deserve judging by the standards of numerical analysis — order, stability and cost — not image statistics alone.

01

Mathematical Formulation & Model I/O

Feedback addressed: clarify the continuous ODE mathematics behind the diffusion model, and specify what inputs are fed to the model and what outputs are generated.

Feedback addressed. *“Clarify the continuous ODE mathematics behind the diffusion model, and specify what is fed to the model and what it outputs.”* — covered here and on the next two slides.

STEP 1 · NOISE INJECTION $0 \rightarrow T$

$$d\mathbf{x} = f(t) \mathbf{x} dt + g(t) d\mathbf{w}$$

Linear drift $f(t)$ and diffusion $g(t)$ carry the data to $\mathcal{N}(\mathbf{0}, \mathbf{I})$; $\mathbf{w}$ is Brownian motion. Fixed, no learning required.

STEP 2 · DETERMINISTIC REVERSAL $T \rightarrow 0$

$$\frac{d\mathbf{x}}{dt} = f(t) \mathbf{x} - \frac{1}{2}g(t)^2 \nabla_{\mathbf{x}} \log p_t(\mathbf{x})$$

Shares exact marginals $p_t(\mathbf{x})$ with the reverse-time SDE (Song et al., 2020) — sampling becomes deterministic, and solvable by classical ODE methods.

Why this reframing matters. With randomness removed, generating an image is numerical integration of a known vector field — accuracy, order and stability now apply directly.

STEP 3 · NEURAL SCORE PARAMETERIZATION

$$\nabla_{\mathbf{x}} \log p_t(\mathbf{x}) \approx - \frac{\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t)}{\sigma_t}$$

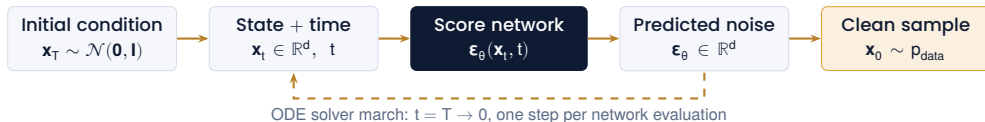
Denoising score matching trains $\boldsymbol{\epsilon}_{\theta}$ to predict the noise; dividing by the schedule σ_t recovers the score.

Substituting yields a semi-linear ODE

$$\frac{d\mathbf{x}}{dt} = \underbrace{f(t) \mathbf{x}}_{\text{stiff, linear, cheap}} + \underbrace{g(t) \boldsymbol{\epsilon}_{\theta}(\mathbf{x}, t)}_{\text{non-linear, one network call}}$$

The split every solver in this study must confront: one term is stiff but analytically tractable, the other dominates the cost.

Mathematical core: sampling is an initial value problem — $\mathbf{x}_0 = \mathbf{x}_T + \int_T^0 [f(t)\mathbf{x} + g(t)\boldsymbol{\epsilon}_{\theta}(\mathbf{x}_t, t)] dt$



Fed to the model

- Gaussian noise $\mathbf{x}_T$ as the initial condition at $t = T$.
- Per step: current state $\mathbf{x}_t \in \mathbb{R}^d$ and scalar time t .

Produced by the model

- Per call: predicted noise $\epsilon_\theta(\mathbf{x}_t, t) \in \mathbb{R}^d$.
- After the march: a clean sample $\mathbf{x}_0 \sim p_{\text{data}}$.

02

Base Paper & The Numerical Core

Lu, Zhou, Bao, Chen, Li & Zhu (Tsinghua University) — *DPM-Solver: A Fast ODE Solver for Diffusion Probabilistic Model Sampling*, NeurIPS 2022, arXiv:2206.00927.

Core innovation

- Solves the stiff linear term $f(t)\mathbf{x}$ **analytically**, by variation of constants.
- Leaves an exponentially weighted integral over $\boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)$ alone.
- Taylor-expands that integral into exponential integrators of order 1, 2 and 3.
- Order 1 reduces exactly to DDIM — Euler's method in disguise.

THE EXPONENTIAL-INTEGRATOR IDEA

$$\mathbf{x}_t = \underbrace{e^{\int_s^t f} \mathbf{x}_s}_{\text{exact linear flow}} + \underbrace{\int_s^t e^{\int_s^\tau f} g \boldsymbol{\epsilon}_\theta d\tau}_{\text{expanded to order } p}$$

Why it matters. Discretization error is confined to the smooth neural term, so accuracy per network call rises sharply — high-quality samples in 10–20 evaluations where DDIM needs 100–250.

Semi-linear ODE → Variation of constants → Weighted integral → Taylor expansion, order p → DPM-Solver-1 / 2 / 3

1 Order never measured

Asymptotic order is proven on paper, but never **numerically measured** against ground-truth ODE trajectories.

2 FID is not error

FID measures image *distribution* statistics — not the discretization error $\|\mathbf{x}_N - \mathbf{x}^*\|_2$.

3 Stiffness unknown

Stability bounds near the stiff boundary $t \rightarrow 0$ are unmeasured, so no step-size guidance exists.

Reproduction target. Recover baseline CIFAR-10 sample quality in 10–20 network evaluations (against 100–250 for DDIM), and supply the layer the paper omits: rigorous ground-truth verification of the **true observed convergence order**.

03

Generalizability: A Multi-Tier Suite

Feedback addressed: if we only test on a single toy problem, how can we be sure the findings hold for the entire problem set?

Feedback addressed. *“The original proposal tested a single toy problem — if we only test on one, how are we sure the findings hold for the entire problem set?”* Answer: the suite below.

1 Stiff linear suite

Anisotropic Gaussians,
 $d = 2 \dots 100$

- Closed-form score;
 $\kappa \in [1, 10^4]$ swept.
- **Target:** stress-test stability limits.

2 Non-linear suite

GMMs & Swiss roll

- Reference:
Dormand–Prince RK8(7),
tol 10^{-14} .
- **Target:** verify order under curvature.

3 Image manifolds

Pretrained CIFAR-10,
 $d = 3072$

- Step budget swept,
 $N = 5 \dots 100$.
- **Target:** confirm speedups hold under neural error.

Property	Tier 1 · Linear	Tier 2 · Non-linear	Tier 3 · Empirical
Score	Closed form	Closed form	Neural network
Dimension	$d = 2 \dots 100$	Low, curved	$d = 3072$
Stiffness	Swept, $\kappa \leq 10^4$	Moderate	Set by the schedule
Ground truth	Analytical	RK8(7), tol 10^{-14}	Fine-grid reference
Primary read-out	Stability envelope	Observed order	Error and FID vs. NFE

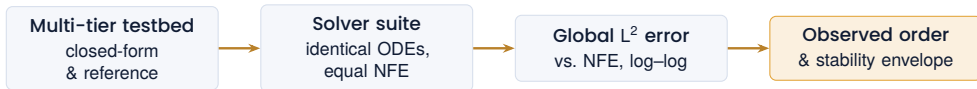
Reading the table. Moving left to right trades exactness of the reference solution for realism of the vector field. A solver claim that survives all three columns is a claim about the **method** — not about one convenient test problem.

Why this generalizes. Convergence order $\mathcal{O}(h^p)$ is governed by vector-field smoothness and Lipschitz constants — exactly the quantities these three tiers parameterize and sweep.

04

Methodology, Solvers & Metrics

Every solver meets every test problem on identical ODEs, at an identical budget of network evaluations.



Solvers implemented (equal NFE budget)

- **Exponential:** DPM-Solver-1, -2, -3
- **Explicit RK:** Euler, Heun (RK2), RK4
- **Adaptive:** RK45, embedded error pair
- **Multistep:** Adams–Bashforth 2 (AB2)

Expected theoretical orders

Method	Order	NFE/step
Euler / DPM-Solver-1	1	1
Heun / DPM-Solver-2	2	2
DPM-Solver-3	3	3
RK4	4	4
Adams–Bashforth 2	2	1

Observed convergence order

Log-log slope of $\|\mathbf{x}_N - \mathbf{x}^*\|_2$ against NFE — measured, not assumed.

Efficiency frontier

Wall-clock time at matched error tolerances — the practical price of an extra order.

Stiffness stability limit

Largest step $h_{\max}$ sustainable before divergence as $t \rightarrow 0$, per solver and per κ .

Empirical FID

Sample quality against NFE on CIFAR-10, reported alongside — never in place of — the L^2 error.

Execution stack: Python · NumPy / SciPy · PyTorch · inference only, no training · GPU for CIFAR-10, CPU for the analytical suite.



What This Project Delivers

- Measured convergence orders for seven solvers on the probability-flow ODE, verified against ground-truth trajectories.
- A stability map of the stiff boundary $t \rightarrow 0$, giving concrete step-size guidance.
- An efficiency frontier that separates genuine solver gains from distribution-level artifacts of FID.